MILITARY SPECIFICATION

TRANSPARENCY, ANIMATED; REDUCING VALVE FOR HIGH TEMPERATURE FLUIDS, GAS-LOADED

This specification has been approved by the Naval Training Device Center, Department of the Navy.

The technical requirements for procuring the animated transparency described herein shall consist of this document and the issue in effect of Specification MIL-T-82300.

1. SCOPE

1.1 This specification covers the requirements for an animated transparency of a gas-loaded type of pressure reducing valve for high temperature fluids.

1.1.1 This item has been designated by the Naval Training Device Center as Device 29HS4 for Navy procurement. (see 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)

Military

MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-T-82300	Transparencies; Animated, General Specification for

FSC-6910

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STANDARDS

Federal

FED-STD-595	Colors

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

821800C90001	A/T, Reducing Valve for High Temperature Fluids, Gas-Loaded - General Arrangement

PHOTOGRAPHS

Naval Training Device Center (NAVTRADEVCEN)

BXA-13728-5-69 through BXA-13730-5-69	Gas-Loaded Reducing Valve for High Temperature Fluids, Device 29HS4

PUBLICATIONS

Naval Training Device Center (NAVTRADEVCEN)

NAVEXOS P-1704	Guide for Animated Transparency, Gas-Loaded Reducing Valve for High Temperature Fluids, Device 29HS4

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with 3.1.1 through 3.1.3. Materials not specified in 3.1.1 through 3.1.3 shall be in accordance with the materials requirements of MIL-T-82300.

3.1.1 Cements.- Cements used for joining plastic components shall be those cements recommended by the manufacturer of the plastic and tested by the contractor as suitable for meeting the cementing and performance requirements of this specification.

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3.1.2 Metals.- Except for springs and actuating arm, all metal parts and fastenings used in the device shall be stainless steel or be protectively plated in accordance with QQ-C-320. The aluminum actuating arm shall be clear anodized in accordance with MIL-A-8625.

3.1.3 Wood and wood products.- Materials for the top and bottom of the carrying case shall be 1/4 inch thick exterior grade plywood, smooth two sides.

3.2 Design.- The animated transparency shall be designed in accordance with the design requirements of MIL-T-82300 and shall be capable of demonstrating all the utilization requirements of NAVEXOS P-1704 except the numerical sequence of the controls. The numerical sequence of the controls shall be as shown on NAVTRADEVCEN drawing 821800C90001.

3.3 Construction.- The animated transparency shall be constructed in accordance with the drawing of 3.2. Construction details not specified on the drawing of 3.2 shall be as shown on NAVTRADEVCEN photographs BXA-13728-5-69 through BXA-13730-5-69. Construction details, not specified in 3.3 through 3.3.2 shall be in accordance with the construction requirements of MIL-T-82300.

3.3.1 Cementing.- Cementing shall be limited to those areas where assembly with screws is clearly impractical. Cemented joints shall be free of air bubbles and stress-crazing and shall have the transparency of the original plastic throughout the joint. The surfaces to be cemented shall be thoroughly softened by means of a solvent, so that a soft layer or cushion is produced on one surface and the subsequent uniting and hardening will effect a strong homogeneous bond. Bonding by capillary attraction shall require specific written approval of the procuring activity for each specific variance. For recommendations on cementing acrylic plastics, the publications cited in the cementing requirements of MIL-T-82300 may be used.

3.3.2 Focusing mark.- The focusing mark shown on the drawing of 3.2 shall have the projected image of the cross hairs and the 1/4-inch diameter opening in the mask concentric. The cross hairs shall be placed at a level to result in optimum focus of all projected components of the device.

3.4 Performance.- The animated transparency, when used on the overhead projector, shall withstand the temperatures produced by a 1,000-watt projection lamp for a period of 45 minutes with the projector's heat filter removed, in a room with ambient temperatures ranging to 90 degrees Fahrenheit. At the end of this period, the animated transparency shall meet the conditions listed in the performance requirements of MIL-T-82300.

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3.5 Details of components.-

3.5.1 Baseplate.- Except as specified in 3.5.1.1, the baseplate shall be in accordance with the baseplate requirements of MIL-T-82300.

3.5.1.1 Dust seal.- The parting line between the layers of the baseplate shall be sealed with white opaque polyester film so as to be dustproof and to mask edge lighting effects. Dust seals shall be provided wherever a laminated component or flap is used. Such seals, in projectable areas, shall be applied to the edges only.

3.5.2 Movable components.- Movable components shall be in accordance with 3.5.2.1 and 3.5.2.2. Components not specified in 3.5.2.1 and 3.5.2.2 shall be in accordance with the movable components requirements of MIL-T-82300.

3.5.2.1 Overall height.- The maximum thickness of the sum of those components which are projected shall not exceed 3/4-inch from the top of the baseplate to the top plane of the highest component. The projected image shall be sharp and clear. Projectable movable components shall have a vertical separation to prevent abrasion of the moving parts. part This separation shall not be less than 1/32-inch during the operating test of MIL-T-82300.

3.5.2.2 Fasteners.- Screws shall be used for the assembly of moving parts and shall be secured to prevent loosening by rotation or movement of the parts. Threaded portions of screws shall not be used as bearing surfaces for moving parts. Rivets shall not be used without prior approval in writing from the procuring activity.

3.5.2.2.1 Selection of screws.- The minimum practical number of screws and the smallest practical head diameter of screws for fastening components shall be used within the projected image area.

3.5.3 Stops.- Stops for flaps and slides shall be provided as specified in the stops requirements of MIL-T-82300.

3.5.4 Controls.- Controls for actuating the various components shall be similar to those shown on the drawing of 3.2 to meet the utilization requirements of NAVEXOS P-1704 of 3.2. All controls shall be movable by light finger pressure.

3.5.5 Carrying case.- The carrying case shall be in accordance with the carrying case requirements of MIL-T-82300. Materials not specified in Figure 2 of MIL-T-82300 shall be as specified in 3.1.3 of this specification.

3.6 Dimensions and tolerances.- Dimensions and tolerances shall be in accordance with the dimensions and tolerance requirements of MIL-T-82300.

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3.7 Finish and color.-

3.7.1 Finish.- Finish shall be in accordance with the finish requirements of MIL-T-82300.

3.7.2 Color.- Colors for projectable components shall be as specified on the drawing of 3.2.

3.7.2.1 Colors listed in Table I of MIL-T-82300, or equivalent in hue, clarity, transparency and durability as its corresponding commercial acrylic color, shall be used.

3.7.2.2 The carrying case shown in Figure 2 of MIL-T-82300 shall be finished with satin-finish lacquer color gray No. 26152 of FED-STD-595.

3.7.3 Cleaning.- Cleaning shall be in accordance with the cleaning requirements of MIL-T-82300.

3.7.4 Waxing.- Waxing shall be in accordance with the wax requirements of MIL-T-82300.

3.8 Nameplates or product markings.- Except as specified in 3.8.1 and 3.8.2, nameplates or product markings shall be in accordance with the nameplates and product markings requirements of MIL-T-82300.

3.8.1 Nameplate.- A nameplate shall be affixed to each animated transparency in the location specified on the drawing of 3.2.

3.8.2 Product markings.- Product markings, nomenclature and notes shall be as shown on the drawing of 3.2.

3.9 Workmanship.- Workmanship shall be in accordance with the workmanship requirements of MIL-T-82300.

4. QUALITY ASSURANCE PROVISIONS

4.1 Except as specified in 4.1.1, quality assurance provisions shall be in accordance with the quality assurance provisions of MIL-T-82300.

4.1.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the preparation for delivery requirements of MIL-T-82300.

6. NOTES

6.1 Except as specified in 6.2, the notes of MIL-T-82300 shall be applicable.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Procuring activity designation (see 1.1.1)

(c) Selection of applicable levels of preservation, packaging and packing

(d) Gross Weight of shipping containers

(e) Government-furnished and Government-loaned property.

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N207

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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DEPARTMENT OF THE NAVY

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Commanding Officer and Director
Naval Training Device Center
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